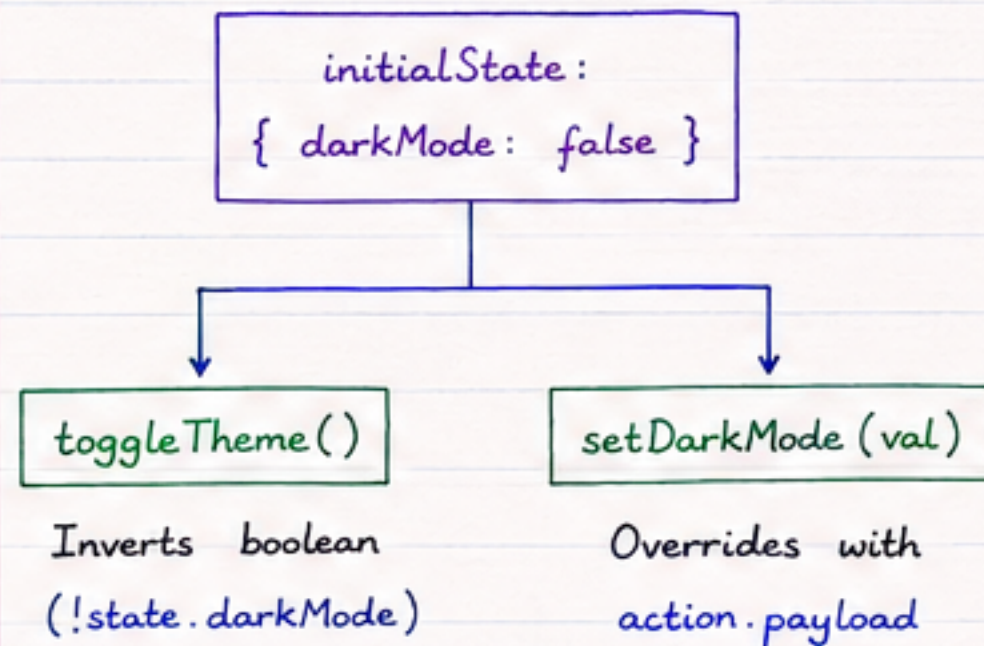


# REDUX TOOLKIT (RTK) - THEME SLICE (themeSlice.js)

## 1. PURPOSE & OVERVIEW

- Role: Manages global UI theme state across the entire React app.
- Goal: Toggle between Dark Mode (true) and Light Mode (false), or explicitly assign a specific boolean state via payload.



### MENTAL MODEL

Your UI doesn't store theme.  
It asks Redux →  
"What is the current theme?"  
Redux answers →  
"darkMode is true/false."  
UI updates accordingly!



## 2. ANATOMY OF 'themeSlice.js'

createSlice() combines:

- ★ Slice Name: "theme"  
(used as action prefix,  
e.g., `theme/toggleTheme`)
- ★ Initial State: { darkMode: false }
- ★ Reducers: Case functions that update the state using Immer.js.

## 3. CODE IMPLEMENTATION

```
import { createSlice } from "@reduxjs/toolkit";

const initialState = {
  darkMode: false, // Default is Light Mode
};

const themeSlice = createSlice({
  name: "theme",
  initialState,
  reducers: {
    // Action 1: Toggle current boolean value
    toggleTheme: (state) => {
      state.darkMode = !state.darkMode;
    },
    // Action 2: Directly set to a specific boolean (true/false)
    setDarkMode: (state, action) => {
      state.darkMode = action.payload;
    },
  },
});

// Export Action Creators (for dispatching in UI)
export const { toggleTheme, setDarkMode } = themeSlice.actions;

// Export Reducer (for store registration)
export default themeSlice.reducer;
```

## 4. INTEGRATION CHECKLIST (Store → UI → DOM)

### Step 1: Register in 'store.js'

```
reducer: {
  theme: themeReducer // State path: state.theme.darkMode
}
```

### Step 2: Read & Dispatch in Component (ThemeToggle.jsx)

```
const darkMode = useSelector((state) => state.theme.darkMode);
const dispatch = useDispatch();
<button onClick={() => dispatch(toggleTheme())}> Toggle </button>
```

### Step 3: Sync to HTML for Tailwind (App.jsx)

```
useEffect(() => {
  document.documentElement.classList.toggle("dark", darkMode);
}, [darkMode]);
```

## 5. COMMON PITFALLS & CHEAT SHEET

- ★ Redux Toolkit Uses Immer:  
Direct mutations like `state.darkMode = ...` are completely safe inside `createSlice` reducers.
- ★ Action Type Strings:
  - `dispatch(toggleTheme())`  
→ { type: 'theme/toggleTheme' }
  - `dispatch(setDarkMode(true))`  
→ { type: 'theme/setDarkMode', payload: true }
- ★ Tailwind Class Strategy:  
Don't forget `'darkMode: class'` (or `'selector'`) in your CSS configuration for the `'dark:'` utility variants to trigger.

### QUICK RECAP

#### ACTIONS

```
toggleTheme()
  → Flip (true ↔ false)

setDarkMode(val)
  → Set to specific value
  (true/false)
```

#### STATE PATH

`state.theme.darkMode`

Single source of truth  
for theme in the app!

